Document Control

Title **Diagnosis and Management of Fungal Infections in Critical Care Guideline**			
Author Dr Tom Lewis Samantha Dymond		**Author's job title** Consultant Medical Microbiologist Antibiotic Pharmacist	
Directorate Planned Care		**Department** Microbiology	
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Main Contact Samantha Dymond Microbiology Department North Devon District Hospital Raleigh Park Barnstaple, EX31 4JB		**Tel: Direct Dial** – 01271 322798 **Tel: Internal** – 2798 **Email**: samantha.dymond@nhs.net	
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Superseded Documents			
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1. Purpose

1.1. This document sets out Northern Devon Healthcare NHS Trust's best practice guidelines for appropriate microbiological investigation and antimicrobial prescribing in adult patients with suspected or confirmed fungal infection on the critical care unit.

1.2. This guideline applies to all adults and must be adhered to. Special considerations exist for pregnant and breastfeeding patients; liaise with specialist clinicians as appropriate in these cases. See separate guidance for paediatric patients.

1.3. Non-compliance with this guideline may be for valid clinical reasons only. The reason(s) for non-compliance must be documented clearly in the patient's notes.

1.4. This guideline is primarily aimed at all prescribing teams but other staff (e.g. nursing staff, pharmacists) may need to familiarise themselves with some aspects of the guideline.

1.5. Implementation of this guideline will ensure that:

- Fungal infection is managed according to current evidence and standards of practice in the wider healthcare community.
- A standard of care is specified to facilitate a consistent approach between critical care, microbiology and pharmacy in terms of patient management, specimen processing and drug availability.

2. Responsibilities

2.1. Responsibility for education and training lies with the Lead Consultant Microbiologist for Antibiotic Stewardship. It will be provided through formal study days and informal training on the ward.

2.2. The author will be responsible for ensuring the guidelines are reviewed and revisions approved by the Clinical Audit and Guidelines Group in accordance with the Document Control Report.

2.3. All versions of these guidelines will be archived in electronic format by the author within the Antibiotic Stewardship policy archive.

2.4. Any revisions to the final document will be recorded on the Document Control Report.

2.5. To obtain a copy of the archived guidelines, contact should be made with the author.

2.6. Monitoring of implementation, effectiveness and compliance with these guidelines will be the responsibility of the Lead Clinician for Antibiotic Stewardship. Where non-compliance is found, the reasons for this must have been documented in the patient's medical notes.

Role of Antibiotic Working Group (AWG)

2.7. The AWG is responsible for:

- Leading antimicrobial guideline development and review within Northern Devon Healthcare Trust
- Involving all relevant stakeholders in guideline development and review

3. Contacts

3.1. Contact numbers:

- Microbiologist — Bleep 193. Via switchboard out of hours.
- Antibiotic Pharmacist — Bleep 029 (Mon-Fri only)

4. Diagnosis and Management of Fungal infections in Critical Care

4.1. See appendix 1

5. Monitoring Compliance with and the Effectiveness of the Guideline

Suggested audit criteria

5.1. The following could be used:

- Percentage of patients with invasive fungal infection who undergo appropriate fungal testing
- Percentage of patients with invasive fungal infection who receive treatment as per guidelines

Process for Implementation and Monitoring Compliance and Effectiveness

5.2. Incidents involving fungal infection should be reported according to the Trust's Incident Reporting Policy. Critical incident reports relating to fungal infection will be collated by the Antibiotic Pharmacist. Results will be reported on an annual basis to the Clinical Audit and Guidelines Group.

6. Equality Impact Assessment

6.1. The author must include the Equality Impact Assessment Table and identify whether the policy has a positive or negative impact on any of the groups listed. The Author must make comment on how the policy makes this impact.

Table 1: Equality impact Assessment

Group	Positive Impact	Negative Impact	No Impact	**Comment**
Age	X			Separate guidance for paediatrics
Disability			X	
Gender			X	
Gender Reassignment			X	
Human Rights (rights to privacy, dignity, liberty and non-degrading treatment)			X	
Marriage and civil partnership			X	
Pregnancy		X		Some treatment advice may harm the unborn foetus, discuss on a case-by-case basis with Obstetricians and Pharmacy for advice.
Maternity and Breastfeeding		X		Some treatments may be excreted in breast milk. Discuss on a case-by-case basis with Paediatricians and Pharmacy for advice.
Race (ethnic origin)			X	
Religion (or belief)			X	

7. References

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7.2. Ullmann, A. 2016. ESCMID Online Lecture Library: Are the immunosuppressed and other populations exempt from guidelines recommending narrow-spectrum and short-term therapy? FOCUS: FUNGAL DISEASE. [online] https://www.escmid.org/escmid_publications/escmid_elibrary/material/?mid=36139 **samples, treatment options, risk assessment**

7.3. John F. Fisher, Jack D. Sobel, Carol A. Kauffman, Cheryl A. Newman. 2011. *Candida* Urinary Tract Infections – Treatment. Clinical Infectious Diseases, Volume 52, Issue suppl_6, 15 May 2011, Pages S457–S466, https://doi.org/10.1093/cid/cir112 **candida UTI management slides**

7.4. Markus Ruhnke, Stefan Schwartz. 2016. Recent developments in the management of invasive fungal infections in patients with oncohematological diseases. Therapeutic Advances in Hematology. https://doi.org/10.1177/2040620716656381 https://journals.sagepub.com/doi/full/10.1177/2040620716656381 **suggests course lengths for candidaemia brain abscess and hepatosplenic lesions, and candidaemia no source. Suggests steroids for refractory fevers when infection controlled.**

7.5. Frederic Tissot, Samir Agrawal, Livio Pagano, Georgios Petrikkos, Andreas H. Groll, Anna Skiada, Cornelia Lass-Flörl, Thierry Calandra, Claudio Viscoli, Raoul Herbrecht. 2017. ECIL-6 guidelines for the treatment of invasive candidiasis, aspergillosis and mucormycosis in leukemia and hematopoietic stem cell transplant patients. Haematologica March 2017 102: 433-444; doi:10.3324/haematol.2016.152900 http://www.haematologica.org/content/102/3/433 **treatment regimens, monitoring**

7.6. Shahid Husain, Amparo Sole, Barbara D.Alexander, Saima Aslam , Robin Avery, Christian Benden, Eliane M.Billaud, Daniel Chambers, Lara Danziger-Isakov, Savitri Fedson, Kate Gould, Aric Gregson, Paolo Grossi, Denis Hadjiliadis, Peter Hopkins, Me-Linh Luong, Debbie J.E.Marriott, Victor Monforte, Patricia Muñoz, Alessandro C.Pasqualotto, Antonio Roman, Fernanda P.Silveira, Jeffrey Teuteberg, Stephen Weigt, Aimee K.Zaas, Andreas Zuckerman, Orla Morrissey. 2016. ISHLT GUIDELINES: The 2015 International Society for Heart and Lung Transplantation Guidelines for the management of fungal infections in mechanical circulatory support and cardiothoracic organ transplant recipients: Executive summary. Journal of Heart and Lung Transplant [online] via https://www.jhltonline.org/article/S1053-2498(16)00054-1/pdf http://dx.doi.org/10.1016/j.healun.2016.01.007 **suggested regimens for SOT recipients, testing limitations of non-haem/onc patients with GM**

7.7. Center for Disease Control (USA). 2007 (updated 2018). Type and Duration of Precautions Recommended for Selected Infections and Conditions: Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings [online] via https://www.cdc.gov/infectioncontrol/guidelines/isolation/appendix/type-duration-precautions.html#H **isolation guidance for histo/blasto/cocc… infections**

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7.11. Personal correspondence with A. Varey Lead Critical Care Pharmacist RUH, Bath. 2019-2020

7.12. Personal correspondence with N. Powell Lead Antimicrobial Consultant Pharmacist, RCH Treliske, Truro. 2019-2020 **adding in Candida score, discussion re: whether to routinely swab, what samples to test, how often to send B-D-glucan / galactomannan**

7.13. Andrew H. Limper, Kenneth S. Knox, George A. Sarosi, Neil M. Ampel, John E. Bennett, Antonino Catanzaro, Scott F. Davies, William E. Dismukes, Chadi A. Hage, Kieren A. Marr, Christopher H. Mody, John R. Perfect, and David A. Stevens. 2011. An official American Thoracic Society Statement: Treatment of Fungal Infections in Adult Pulmonary and Critical Care Patients. Am J Respir Crit Care Med Vol 183. pp 96–128, DOI: 10.1164/rccm.2008-740ST https://www.thoracic.org/statements/resources/tb-opi/treatment-of-fungal-infections-in-adult-pulmonary-critical-care-and-sleep-medicine.pdf **critical care specific infections and treatment courses**

7.14. Oren, I; Paul, M. 2014. Up to date epidemiology, diagnosis and management of fungal infections. Clinical Microbiology and Infection vol 20 supplement 6, pp1-4. https://www.clinicalmicrobiologyandinfection.com/article/S1198-743X(14)60325-6/fulltext DOI: https://doi.org/10.1111/1469-0691.12642 **prevalence, risk groups for different infections, treatment strategies**

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8. Associated Documentation

- Incident reporting policy
- Antibiotic guidelines for management of severe sepsis and septic shock
- Antimicrobial prescribing policy

9. Appendix

9.1. Name of guideline on app

Fungal Infections in Critical Care

9.2. Location on app

Secondary Care
Infection
Adult treatment
Fungal Infections in Critical Care

9.3. Header

Invasive fungal infection should be suspected in patients who are profoundly immunosuppressed and/or long critical care admission and/or who have rising inflammatory markers with worsening systemic infection symptoms on optimal antibiotic treatment for at least 48 hours. Formal diagnosis relies upon

- Tissue or other clinical sterile specimens yielding fungal growth and appearance of organisms on direct microscopy and staining in the lab
- Radiological evidence of fungal infection in tissues/organs

9.4. Risk Assessing Haem/Onc Patients for prophylaxis requirements

High Risk - mould active prophylaxis	Low Risk - candida prophylaxis	Low Risk - no prophylaxis
Allo-HSCT **Intensive treatment for ALL, AML, MDS** **Significant GVHD** –till resolved. **CML intensive chemo** **Severe aplastic anaemia** Duration Allografts to day 75-100 GVHD – 16 weeks or until prednisolone <10mg OD Others – neutrophil recovery	**Auto-SCT** – candida prophylaxis if mucositis or recent excessive chemo until neutropenia resolved **Myeloma** – fluconazole or no prophylaxis **Lymphoma** - intensive/dose-escalated therapy **Solid tumours** – if profound neutropenia and mucositis expected to last for ≥ 7 days in environments with > 10% risk of invasive Candida infection	**MDS** – not undergoing intensive chemo **CML** (treated with TKIs or conventional treatment) **CLL** No prophylaxis (consider in CLL with prolonged neutropenia (>6 months), elderly, advanced and unresponsive disease) **Lymphoma** - standard chemo **Other myeloproliferative neoplasms**
Unclear		
Autograft – mould-active agent if prior IA, neutropenia >2 weeks expected or prolonged neutropenia prior to HSCT **Allo-HSCT** with expected neutropenia <14 days (II, A) **Aplastic anaemia** - Consider prophylaxis for first months after ATG and after HSCT for as long as neutropenia and/or lymphopenia is present **Allogeneic HSCT** with expected neutropenia >14 days Corticosteroids >1 mg/kg prednisolone equivalent and neutrophils <1 × 10 9 /L for >1 week Corticosteroids >2 mg/kg prednisolone equivalent >2 weeks High-dose cytarabine Fludarabine use in highly treatment-refractory patients with CLL or low-grade lymphoma Alemtuzumab use, especially in highly treatment-refractory patients with CLL or lymphoma		

9.5. Risk Assessing All Patients for Likelihood of Developing IFI

- Invasive *Candidiasis*

  Candida scoring using the table may help to decide if the patient is a candidate for starting early treatment.

Parenteral nutrition	Score 1 if present	
Surgery	Score 1 if present	
Multifocal *Candida* colonisation	Score 1 if present	
Severe sepsis	Score 2 if present	
Total (0-2 low risk; 3-5 high risk):		

Also consider the following factors in all patients:

- Immunosuppression
  - Granulocytopenia
  - Immune modulating drugs (commonly glucocorticoids, chemotherapeutic agents, transplant anti-rejection drugs)
  - Transplant recipients – more common in the initial 30 day period post-organ transplant
  - Poor immune status due to co-morbidities (commonly diabetes, liver failure, malignancy, HIV, hepatitis)

- Breach of normal defensive mechanisms
  - Broad spectrum antibiotics
  - Mucosal barrier injury (mucositis)
  - Burns / large wound sites / extensive debridement of skin
  - GI perforation
  - Catheters (IV, urinary)
  - Haemo-dialysis/-filtration
  - IV drug use (illicit and prescribed)
  - Pancreatitis
  - Prophylaxis failure (interactions, not taken as prescribed, levels too low)
- Length of exposure in risky environment
  - Length of stay in hospital / ICU
  - Duration of surgery
  - Duration of broad spectrum antibiotic therapy (colonization)
  - Severity of underlying illness – APACHE II/III associated with worse outcomes

- Aspergillosis – no recognised risk scoring tool. Consider the following factors:
  - Long periods of immunosuppression
    - Chemotherapy
    - High dose steroids
    - Transplant recipients (organ / stem cell), more commonly after the initial 30 days post-organ transplant
    - HIV/AIDS
    - Other immune deficiency (caused by co-morbidities or primary immune deficiency)
  - Chronic exposure to high environmental levels
    - Pulmonary infection vs. inflammation
  - Acute exposure to high environmental levels whilst unwell ± immunosuppression
    - Building work near home / hospital
- Cryptococcus – no recognised scoring tool. Consider the following factors:
  - Immunosuppression
    - Transplant recipients (organ/stem cell)
    - HIV/AIDS
    - Immune-modulating drugs (DMARDs, chemotherapeutic agents)
    - Other immune deficiency (caused by co-morbidities or primary immune deficiency)
  - Exposure to environmental contamination
    - Soil
    - Bird droppings
- Mucormycosis (zygomycosis) – no recognised scoring tool. Consider the following factors:
  - Similar patient profile to Aspergillosis but diagnosis difficult and biopsy of the lesion required for definitive identification. Some signs which may help to suspect a diagnosis of mucormycosis:

- Sinus disease
- Concomitant diabetes
- Occurs on mould-active prophylaxis/treatment with voriconazole
- Reverse halo sign on chest CT-scan
- Open wound (trauma/burn) which rapidly changes in appearance despite adequate broad spectrum antibiotic cover

- Pneumocystis jirovecii – no recognised scoring tool. Consider the following factors:
  - Not strictly IFI, but important consideration on the ICU for patients with pneumonia and atypical CXR e.g. "ground glass appearance"
  - Long periods of immunosuppression
    - Chemotherapy
    - High dose steroids
    - Transplant recipients (organ / stem cell)
    - HIV/AIDS
    - Other immune deficiency (caused by co-morbidities or primary immune deficiency)
- Other invasive fungal infections – no recognised scoring tool. Consider the following factors:
  - The following are rarely seen in the UK, usually associated with history of foreign travel. These organisms are all highly pathogenic and immunocompetent patients are also affected. Not usually transmitted from person to person, but obtain side-room if possible, especially where draining lesions or wet cough.
    - Histoplasma - Pulmonary infection most commonly seen
    - Blastomycosis - Pulmonary, orthopaedic and skin infection most commonly seen, but occasional CNS infection
    - Coccidioidomycosis - Pulmonary infection most commonly seen, immunosuppressed individuals at risk of disseminated disease and CNS complication

9.6. Diagnosis

Clinical Examination

- Full external inspection of the patient should be performed to identify potential sources of infection from skin / wounds – ensure areas which are prone to moisture lesions or concealed under dressings are also examined.
- Travel history (may be omitted if long admission prior to developing suspected IFI)
- Occupational history (may be omitted if long admission prior to developing suspected IFI)
- Consider previous medical history (see above - "risk assessing all patients…")
- Consider circumstances on the ward as part of the work-up:
  - Prevalence and common types of invasive fungal infection (IFI) on the local critical care unit
    - Moulds
    - *Candida spp.*

- Repatriation of patient or recent repatriation of another patient to same bed space from other critical care unit with known high prevalence of IFI, particularly if *Candida auris* or other highly resistant isolates (liaise with local infection prevention and control team and microbiology if found)

Microbiology

- Blood cultures x 2 sets (i.e. 10mL per bottle, 2 each of aerobic and anerobic bottles), preferably from different lines and taken at different times
- If line source suspected, remove and send line tip for culture
- BAL for patients with a suspected chest source
- Stool specimen if suspected GI source
- Urine specimen, preferably MSU, even if no current clinical suspicion of urinary source, to assess for candiduria. If necessary to send CSU, take from freshly inserted catheter
- Wound exudate captured in bottle (preferred), or swab if insufficient loose material, for suspected skin/wound source (contamination potential from cellulose/cotton swabs)
- Lumbar puncture if suspected CNS source, also send parallel serum sample. Request Cryptococcal antigen testing (CRAG) on both CSF and serum samples
- Skin / nail scrape if fungal nail infections and dermatophyte infections of the skin folds, lower legs and feet unresponsive to topical empirical therapy, or if felt to be clinically significant in context of suspected IFI

Histology

- All fluid and tissue samples for direct staining and microscopy

Radiological Imaging

- High resolution CT-scan is sufficient for diagnosis of invasive fungal infection in most deep tissues / organs
- MRI is considered superior for diagnosis of CNS, sinus, ophthalmic, hepatosplenic and gastrointestinal fungal infections, where available, and more suitable for use where concerns about radiation exposure
- Ultrasound and X-ray can sometimes be used (discuss options with local radiology) but these images are more difficult to interpret and diagnosis cannot always be formally confirmed. They may be used to guide treatment escalation / de-escalation if no other scan facility is available on the critical care unit
- Interventional radiology may be required for obtaining organ/deep tissue biopsy

Endoscopic Methods

- Bronchoscopy with BAL for investigation and sampling of suspected pulmonary infiltrates
- OGD with biopsy for patients with signs and symptoms of oesophagitis
- Endoscopy of the lower gastrointestinal tract may be considered if indicated in individual cases e.g. with severe diarrhoea

Biopsy

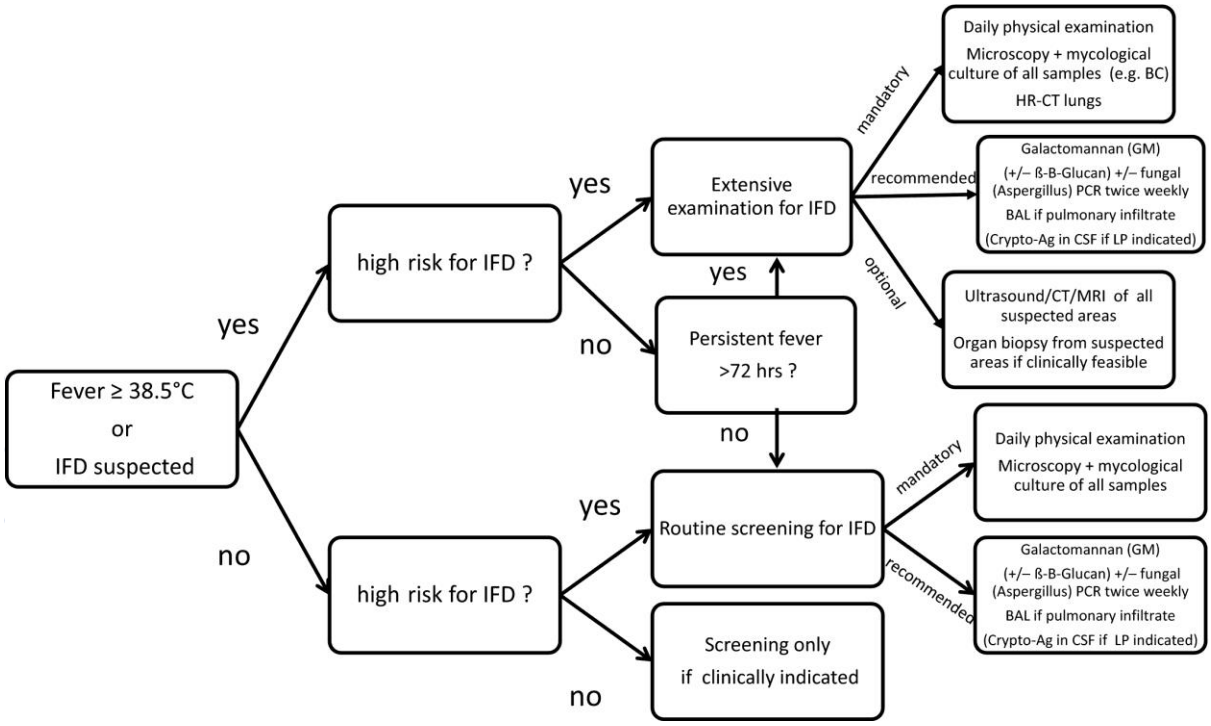
- Specimens should be taken from areas of suspected infection identified by scans, if clinically feasible
- Immunofluorescence (where available) can be used to identify *Pneumocystis* infection, but PCR more usually used in reference labs serving local DGHs.

Serology

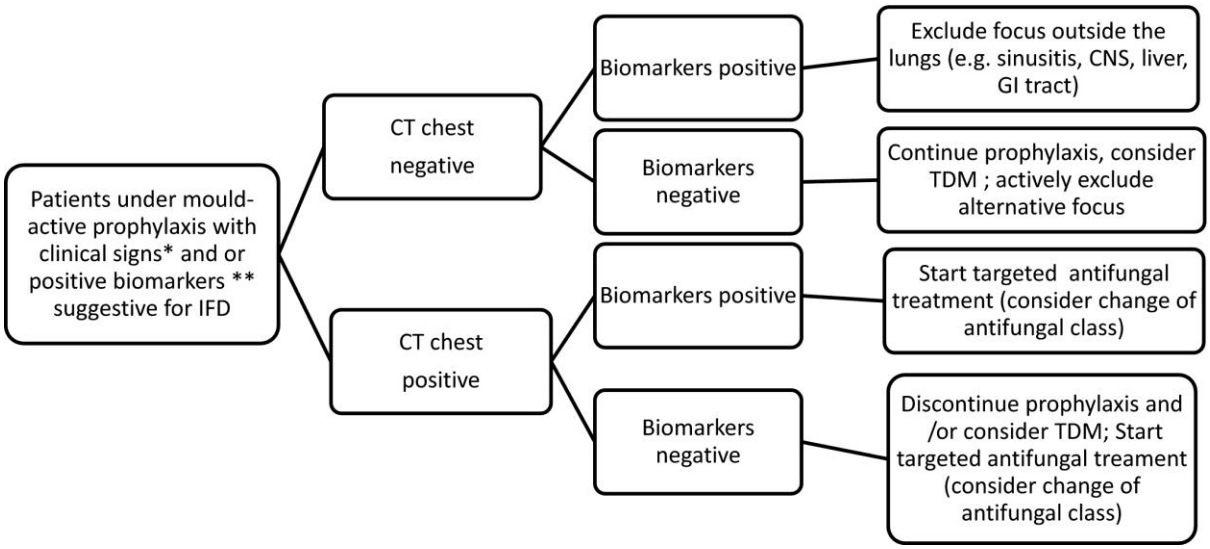
- NB. Interpretation of blood tests for fungal infection (other than directly grown cultures) is challenging and must be undertaken in combination with clinical features and radiological evidence –
- Galactomannan – *Aspergillus* antigen – negative predictive value high, positive predictive value low.
  - Mould-active antifungal prophylaxis reduces the value of a negative test
  - False positives seen with administration of Plasma-Lyte® infusion, some generic beta-lactam antibiotics (co-amoxiclav and piperacillin-tazobactam), transfusions and ingestion of "ice-pops."
  - Serial screening tests over time may yield more valuable clinical guidance than isolated samples for "high-risk" patients who do not currently require treatment. Unexpected high/positive results should be interpreted in the context of clinical symptoms and radiological results due to the propensity for contamination.
  - Patients on mould-active prophylaxis should have BAL samples sent, serum samples not usually recommended unless clinical suspicion of new IFI
  - Patients not on mould-active prophylaxis/treatment should have BAL and serum samples sent for screening
  - Levels thought to correspond to fungal burden, therefore serial tests when the patient is on treatment for IFI may be useful to monitor treatment efficacy.
  - Reports of low sensitivity in non-neutropenic patients (as low as 22%) for diagnosing invasive Aspergillosis. Higher reliance on clinical, histology and radiology diagnosis recommended in these patients.
- ß-D-glucan – non-specific test for cell wall polysaccharides in some yeasts and moulds, **note important exceptions below**.
  - Mucormycosis and *Cryptococcus* infections do not produce ß-D-glucan, **negative results should not be used to guide treatment decisions**
  - Positive in *Candida, Aspergillus, Pneumocystis* infections
  - False positives: diet rich in cereals, cotton/cellulose swabs/gauze/garments in contact with blood or sample site, dialysis membrane, albumin infusion, immunoglobulin infusion, cephalosporin and carbapenem antibiotics.
  - Serial tests over time may yield more valuable clinical guidance than isolated samples. Unexpected high/positive results should be interpreted in the context of clinical symptoms and radiological results due to the propensity for contamination.
- DNA PCR – needs to be interpreted with caution.
  - Microbiology can give advice about when it is most useful to request, except:
    - Suspected PCP (*Pneumocystis jirovecii* pneumonia) - always send respiratory sample for PCR in presence of clinical signs and symptoms: BAL (preferred), if unobtainable, induced deep sputum

- Positive DNA PCR on samples of tissue, catheter urine, exudate/pus may be evidence of colonisation or contamination, rather than infection.
- Exceptions to note:
  - *Pneumocystis jirovecii* PCR on sputum samples, if positive in combination with clinical signs, should be interpreted as active infection
  - Fungal DNA found in blood samples is usually taken to indicate infection requiring treatment in patients whose fever persists despite broad spectrum antibiotic treatment.
  - *Candida* found in mid-stream urine may be considered significant if no other source of infection identified, recent uro-/gynaeco-logical surgery, indwelling stents, or urinary symptoms predominate

9.7. Clinical Assessment – not currently on mould-active antifungal prophylaxis



9.11. Clinical Assessment – currently on mould-active antifungal prophylaxis



9.12. Watch Out For…

- Patients repatriated from other centres where known multi-drug resistance e.g. *Candida auris*
- Seeding of infection from other sites into the eye and the heart – prompt referral on suspicion of IFI to:
  - Ophthalmology for urgent review with fundoscopy,
  - Cardiology for echocardiogram to rule out infective endocarditis
- Antifungals must be monitored closely – known to cause serious adverse effects on:
  - Other drug activity. Some common examples to bear in mind *[list not exhaustive]*:
    - HIV antiretrovirals
    - Some chemotherapeutic agents
    - Organ rejection / transplant agents
    - Anti-epileptics
    - Statins
    - Benzodiazepines
  - Organ systems. Some of the most severely affected listed below – see individual drug monographs for full details of drugs and their side effects
    - Liver toxicity
    - Cardiac toxicity
    - Renal toxicity
    - Haematological toxicity
    - Ocular toxicity
    - Skin reactions (severe)
    - CNS toxicity
- TDM is required for some antifungals, most notably those routinely used for haem/onc/post-transplant prophylaxis:
  - Posaconazole
  - Itraconazole

- Extended course length often required for treatment of IFI – ensure prescriptions clearly state drug not to be stopped before a suitable length of time has elapsed (barring adverse effects)

9.13. Treatment [closed]

Active treatment in response to clinical signs, alongside the radiology, serology and histology work-up: Standard treatment modality for acute patients, or those who unexpectedly go on to develop IFI during admission.

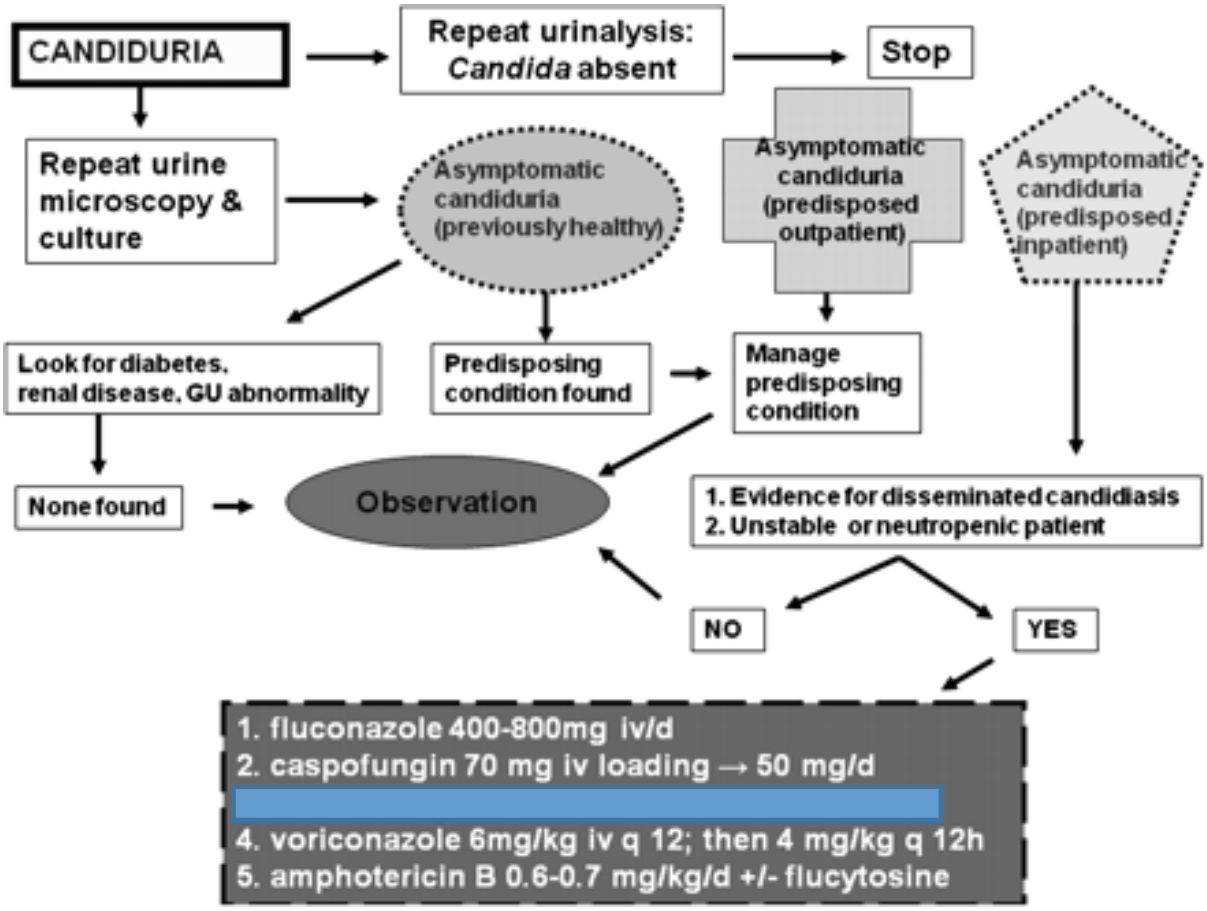
Pre-emptive treatment commenced before the patient becomes clinically unwell, in response to radiological, serological or histological signs developing: Established practice in profoundly immunosuppressed long-term patients for whom serial monitoring is feasible. Rationale to reduce treatment-related toxicity, costs, resistance and improve patient outcomes.

- ## Candidaemia with no organ involvement
  - Initially:
    - Treat with Caspofungin (or liposomal formulation of amphotericin B [less preferred]), pending species and sensitivities
    - If possible, remove/change CVC, other lines and devices which may be potential sources
  - If *Candida spp.* sensitive to azoles and CVC/lines have been changed:
    - Fluconazole via IV route for at least 10 days initially, then if patient well…
    - …switch to PO fluconazole, continue for 14 days after the end of candidaemia (blood cultures to be taken at intervals during treatment – suggested interval daily as per ESCMID guideline)
  - If CVC/lines cannot be changed:
    - Continue Caspofungin / liposomal amphotericin B or suitable other agent as per sensitivities at least until line can be changed, in order to treat any associated biofilm (azoles less effective vs. biofilms) and then until asymptomatic / for 10 days - whichever is longer…
    - …Review sensitivities, switch agent if appropriate and continue treatment for 14 days after the end of candidaemia
  - If *Candida spp.* resistant to azoles
    - Treat according to sensitivities taking into account local formulary preference, for at least 10 days initially, continue for 14 days after the end of candidaemia (blood cultures to be taken at intervals during treatment – suggested interval daily as per ESCMID guideline)
- ## Candidaemia with organ involvement
  - Treat initially as per “no organ involvement” and await species with sensitivities
  - Attempt resection / drainage for source control, if feasible
  - Hepatosplenic lesions – continue treatment until radiological resolution
  - Brain abscesses – continue at least 4 weeks after resolution of symptoms

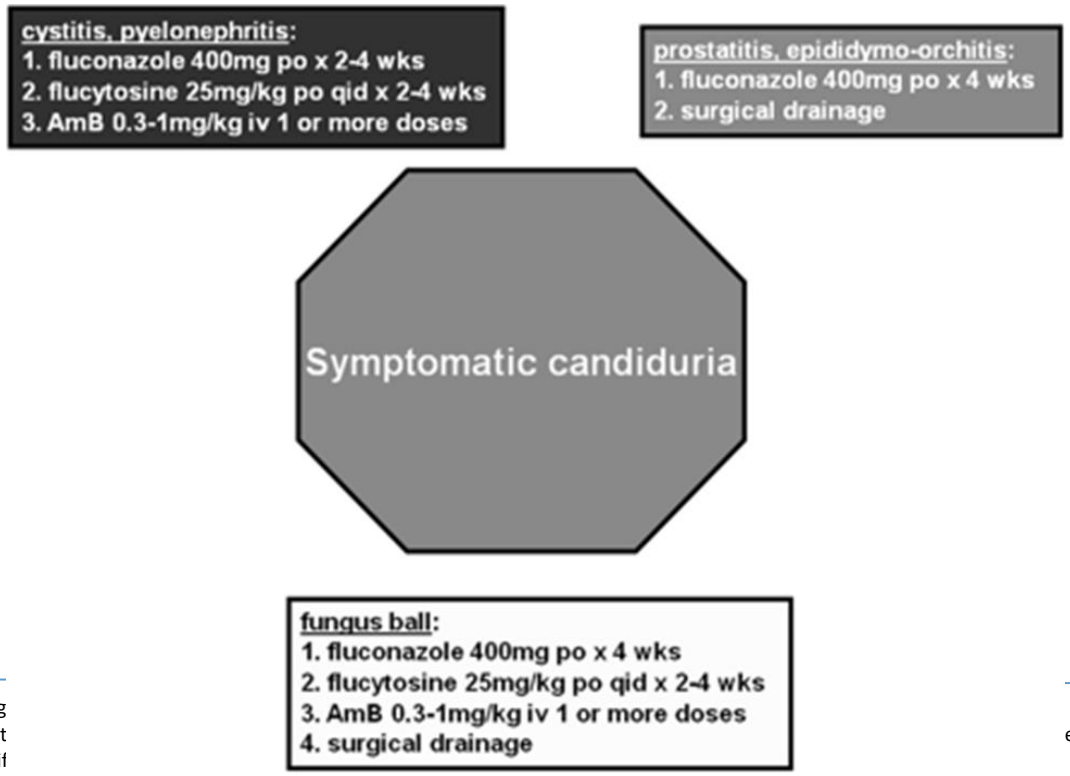
- Some patients may continue to suffer refractory fevers caused by immune reconstitution – providing the infection is currently controlled or has been treated, a trial of corticosteroid could be used to manage symptoms

Candida urinary tract infection

- Management of asymptomatic candiduria



- Management of symptomatic candiduria



- ## Invasive Aspergillosis
  - Initially:
    - If possible, source control according to site of infection by drainage of collections, removal or change of lines/devices
    - First Line: Voriconazole / Isavuconazole IV, continued until clinical and radiographic resolution.
    - Second Line: Liposomal Amphotericin B IV, continued until clinical and radiographic resolution
    - For infections around permanent implanted devices, at least 6-8 weeks initial treatment-dose therapy before review to step down to long-term prophylaxis, even if clinical and radiographic resolution occurs beforehand.
  - Salvage Therapy if first and second-line failure or intolerance: Review sensitivities if organism isolated. Consider echinocandin, other azole, or a combination strategy of azole with echinocandin. NB. Echinocandins not recommended for treating urinary tract infections due to poor penetration.
  - Some patients who continue to be "at risk" due to immunosuppression or continued exposure, e.g. repeated isolation of *Aspergillus* from lung biopsies post-treatment, may require extended treatment, or prophylaxis.
    - Voriconazole particularly, and also other azoles should only be prescribed for outpatients after thorough counselling due to a number of serious side effects – including risk of skin cancer, liver failure, ophthalmic toxicity and heart failure. Patients require a warning card to be issued with their prescription.
    - Arrange with the patient and/or their carers the frequency of follow-up and what to do in case of recurrence of symptoms
- ## Mucormycosis
  - First line: Liposomal Amphotericin B
  - Second line: Posaconazole / Isavuconazole
  - Surgical debridement or resection of infected tissues
  - If applicable: control underlying immune condition
- ## Cryptococcosis
  - Individuals presenting with untreated HIV/AIDS should usually delay commencing treatment with antiretrovirals until 8-10 weeks after starting treatment for cryptococcosis to avoid immune reconstitution syndrome (IRS). For patients who go on to develop IRS, steroids may be used to manage this, as long as effective antifungal treatment continues.
  - Meningitis:
    - Attempt source control if feasible by draining collections, changing lines / devices
    - Combination therapy with: Liposomal Amphotericin B plus flucytosine for 2 weeks, followed by fluconazole or itraconazole 10 wk

- ⇨ OR combination therapy with: Liposomal Amphotericin B plus flucytosine for 6-10 weeks
- ⇨ NB. Ability to use flucytosine may be limited by low platelet counts or neutropenia
- ⇨ Management of intra-cranial pressures key to avoiding complications, acetazolamide and diuretics not recommended to achieve this
- ⇨ Use of steroids associated with worse outcomes in HIV patients, but evidence for sight-sparing in *C.gattii* infection and useful in management of immune reconstitution syndromes

➢ Pulmonary Infections:

- ⇨ Immunocompetent, asymptomatic hosts with *C. neoformans* isolated from respiratory samples – no treatment recommended
- ⇨ Immunocompetent hosts with clinical pulmonary *Cryptococcus* infection: fluconazole / itraconazole for 6 months, or as per sensitivities. Longer course may be required for fluconazole if *C. gattii* due to innate lower sensitivity.
- ⇨ Immune compromise or pulmonary-disseminated infection: treat as per meningitis

- **Other fungal infections – seek microbiology advice in all cases**

➢ Histoplasma

- ⇨ Usually responds to itraconazole. Try liposomal amphotericin B if very unwell. Steroids may be given to alleviate symptoms only in combination with antifungals. Suggested up to 24 month course

➢ Blastomycosis

- ⇨ Usually responds to itraconazole. CNS infection requires treatment with amphotericin B. Suggested at least 12 months for immune suppressed and indefinite for AIDS without immune reconstitution

➢ Coccidioidomycosis

- ⇨ Usually responds to fluconazole, itraconazole. Amphotericin B used in serious infection and meningitis. Suggested at least 12 months treatment for serious pulmonary and lifelong for CNS.

9.14. If No Better… [closed]

- Review cultures and PCR results (where applicable)
- Discuss with microbiology, radiology and parent specialty regarding further investigations, treatment options.
- Referral to alternative centre of care may be appropriate for further management
- Be aware that the course of illness in invasive fungal infection is long and treated very differently to bacterial infections.

9.15. Other Relevant Guidelines [closed]

9.16. Organisms and Sensitivities [closed]

9.17. Version Control [closed]